

Exercises Tutorial 9 (23-05-2024)

Exercise 1

It is hoped that a newly developed pain reliever will more quickly produce perceptible reduction in pain to patients after minor surgeries than a standard pain reliever. The standard pain reliever is known to bring relief in an average of 3.5 minutes with standard deviation 2.1 minutes. To test whether the new pain reliever works more quickly than the standard one, 50 patients with minor surgeries were given the new pain reliever and their times to relief were recorded. The experiment yielded a sample mean of 3.1 minutes and a sample standard deviation of 1.5 minutes. Is there sufficient evidence to indicate, at a 5% level of significance, that the newly developed pain reliever does deliver perceptible relief more quickly?

Exercise 2

In the past the average length of an outgoing telephone call from a business office has been 143 seconds. A manager wishes to check whether that average has decreased after the introduction of policy changes. A sample of 100 telephone calls produced a mean of 133 seconds, with a standard deviation of 35 seconds. Perform the relevant test at the 1% level of significance.

Exercise 3

The average household size in a certain region several years ago was 3.14 persons. A sociologist wishes to test, at the 5% level of significance, whether it is different now. Perform the test using the information collected by the sociologist: in a random sample of 75 households, the average size was 2.98 persons, with sample standard deviation of 0.82 person.

Exercise 4

The recommended daily calorie intake for teenage girls is 2,200 calories/day. A nutritionist at a state university believes the average daily caloric intake of girls in that state to be lower. Test that hypothesis, at the 5% level of significance, against the null hypothesis that the population average is 2200 calories/day using the sample data of size 36, mean 2150, standard deviation 203.

Exercise 5

Suppose that a delivery company claims that they deliver their packages in 2 days on average, and you suspect it's longer than that. To test this claim, you take a random sample of 10 packages and record their delivery times. You find the sample mean is $\bar{x} = 2.3$ days, and the sample standard deviation is 0.35. Do you have sufficient data to show that the company delivers their packages in more than 2 days, considering significance level 0.05? And what assumptions do you need?

Exercise 6

An old research has determined the mean age of a certain population to be 70.65 with a 95 % confidence interval of (67.2, 74.1). A new sample yields $\bar{x} = 75.32$.

- a) What is your conclusion concerning the $H_0 : \mu = 70.65$ with $\alpha = 0.05$?
- b) Is this a one-sided or a two-sided test?

Exercise 7

The hypothesis $H_0 : \mu = 15$ is tested against $H_1 : \mu \neq 15$ with $\alpha = 0.05$. A random sample has following properties: $n = 20$, $\bar{x} = 17.5$, $s = 5.9$.

- a) What assumptions should be made and which distribution should be used? Why?
- b) What is the range of the P-value that you can find with table A4?
- c) What is your conclusion regarding the null hypothesis?

Exercise 8

The average brown trout's I.Q. is 4. Suppose you don't believe this claim. You catch 12 brown trout. A fish psychologist determines the I.Q.s as follows: 5; 4; 7; 3; 6; 4; 5; 3; 6; 3; 8; 5. Conduct a hypothesis test of your belief with $\alpha = 0.05$.

Exercise 9

A child psychologist says that the average time that working mothers spend talking to their children is 11 minutes per day. Suppose a random sample of 100 working mothers spend on average 11.5 minutes per day talking to their children. The prior research suggests the population standard deviation is 2.3. Do we have evidence that working mothers spend more than 11 minutes per day by talking to their children? Use $\alpha = 0.05$.

Exercise 10

Consider the previous exercise, but now with a small change: The random sample does not include 100 working mothers, but only 10 working mothers. Will your conclusion change?

Exercise 11

The Achenbach Child Behaviour Checklist is designed so that scores from normal children are Normal with mean $\mu = 50$ and standard deviation $\sigma = 10$. We are given a sample of $n = 5$ children under stress with an average score of $\bar{x} = 56.0$. Is there evidence that children under stress show an abnormal behaviour? You may assume that σ still is known to be equal to 10. Use level of significance 0.05.

Exercise 12

Solve the previous exercise, assuming that we do not know σ , but are given a sample of $n = 5$ children under stress with an average score of $\bar{x} = 56.0$ and standard deviation $s = 8.5$.